

Checking Stoker's wet dam break

Substitution into the one-dimensional shallow water equations

A subscript on a field is a partial derivative. The depth is h , the depth-averaged velocity is u , and g is gravity. The bed is flat, so $z = 0$. This note takes Stoker's wet dam break from the companion note and substitutes it into the shallow water equations.

1. The equations

On a flat frictionless bed,

$$\begin{aligned} h_t + (hu)_x &= 0, \\ u_t + uu_x + gh_x &= 0. \end{aligned} \tag{1}$$

Where $h > 0$ this is the same as the conservation form

$$\begin{aligned} h_t + (hu)_x &= 0, \\ (hu)_t + \left(hu^2 + \frac{gh^2}{2} \right)_x &= 0. \end{aligned} \tag{2}$$

A jump that moves at speed s satisfies the conservation form in the weak sense when the Rankine–Hugoniot relations hold. With states (h_-, u_-) on the left of the jump and (h_+, u_+) on the right,

$$\begin{aligned} s(h_- - h_+) &= h_- u_- - h_+ u_+, \\ s(h_- u_- - h_+ u_+) &= h_- u_-^2 - h_+ u_+^2 + \frac{g}{2}(h_-^2 - h_+^2). \end{aligned} \tag{3}$$

2. The candidate

Let $0 < h_R < h_L$, $c_L = \sqrt{gh_L}$, and $c_R = \sqrt{gh_R}$. The dam is at $x = 0$: at $t = 0$ the velocity is zero, the depth is h_L for $x < 0$, and the depth is h_R for $x > 0$.

The middle depth h_m is the unique number in (h_R, h_L) for which the two expressions

$$u_m = 2(c_L - c_m) = (h_m - h_R) \sqrt{\frac{g(h_m + h_R)}{2h_m h_R}} \tag{4}$$

agree, where $c_m = \sqrt{gh_m}$. Such an h_m exists because, as h runs from h_R to h_L , the rarefaction value $2(c_L - \sqrt{gh})$ falls continuously from $2(c_L - c_R) > 0$ to 0, while the shock value $(h - h_R)\sqrt{g(h + h_R)/(2hh_R)}$ rises continuously from 0. The two graphs cross once.

For $t > 0$ write $\xi = x/t$ and

$$s = \frac{h_m u_m}{h_m - h_R}. \tag{5}$$

The solution has four pieces:

$$(h, u) = \begin{cases} (h_L, 0) & \text{if } \xi < -c_L \\ \left(\frac{1}{9g}(2c_L - \xi)^2, \frac{2}{3}(c_L + \xi) \right) & \text{if } -c_L \leq \xi \leq u_m - c_m \\ (h_m, u_m) & \text{if } u_m - c_m \leq \xi < s \\ (h_R, 0) & \text{if } \xi > s. \end{cases} \tag{6}$$

The right edge of the fan is $u_m - c_m = 2c_L - 3c_m$, which is where Ritter's parabola reaches $c = c_m$.

The fields depend on x and t only through ξ . The figure takes $h_R = h_L/4$ and draws the depth in units of h_L and the velocity in units of c_L , against $x/(c_L t)$. Still water of depth h_L stands to the left of -1 . The rarefaction runs from there to $\xi = u_m - c_m$, the middle state fills the strip up to the shock $\xi = s$, and undisturbed water of depth h_R lies beyond the shock.

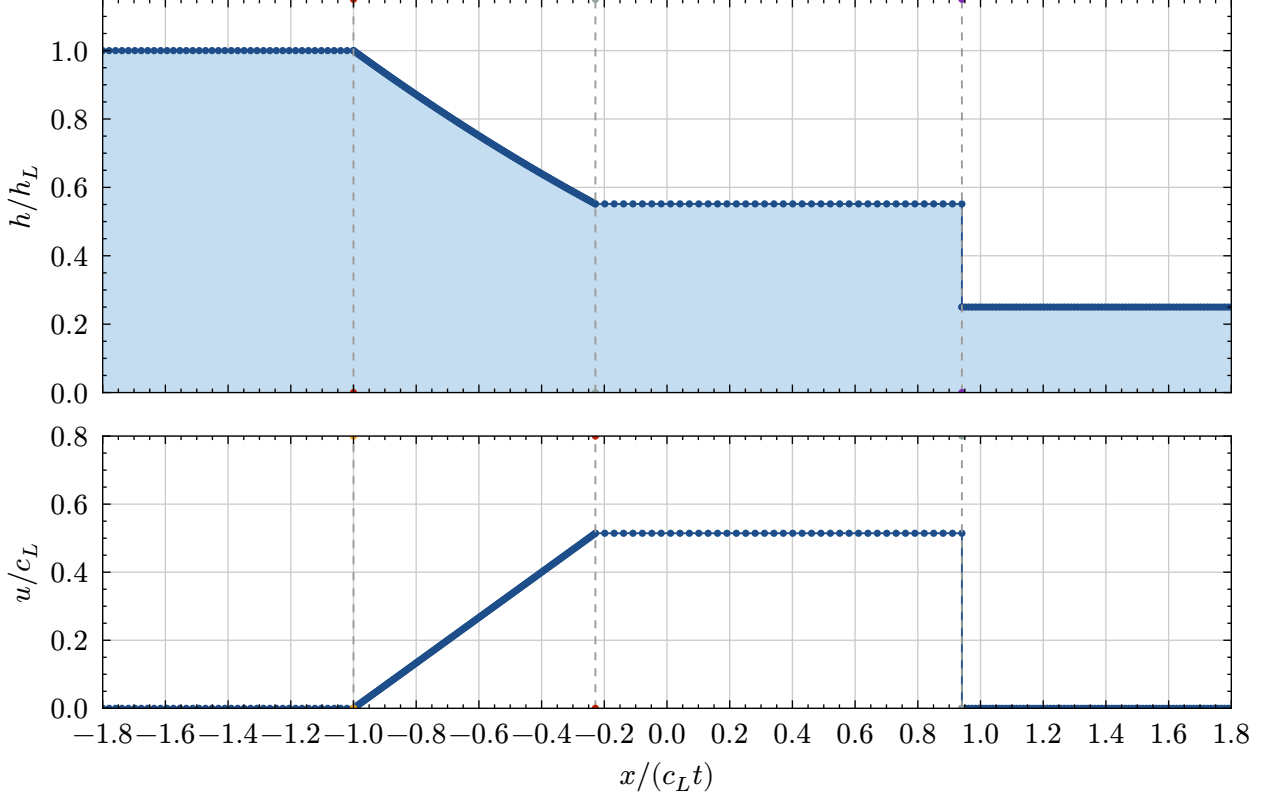


Figure 1: Stoker's solution for $h_R = h_L/4$ at a fixed $t > 0$. The dashed lines are $x = -c_L t$, the downstream edge of the rarefaction, and the shock $x = st$.

3. Still water

For $\xi < -c_L$ the fields are the constants $(h_L, 0)$. For $\xi > s$ they are the constants $(h_R, 0)$. In both regions every derivative vanishes, so both equations reduce to $0 = 0$.

4. The rarefaction

On $-c_L < \xi < u_m - c_m$ the formulas are the same functions of ξ as in Ritter's dry dam break. Set $\varphi = 2c_L - \xi$. Then

$$u = \frac{2}{3}(c_L + \xi), \quad h = \frac{\varphi^2}{9g}, \quad c = \frac{\varphi}{3}, \quad (7)$$

and $u + 2c = 2c_L$. The derivatives are

$$u_t = -\frac{2\xi}{3t}, \quad u_x = \frac{2}{3t}, \quad h_t = \frac{2\varphi\xi}{9gt}, \quad h_x = -\frac{2\varphi}{9gt}. \quad (8)$$

Mass becomes

$$h_t + (hu)_x = \frac{2\varphi}{9gt} \left(\xi - \frac{2}{3}(c_L + \xi) + \frac{2c_L - \xi}{3} \right). \quad (9)$$

The parenthesis is $\xi - (2/3)\xi - (1/3)\xi - (2/3)c_L + (2/3)c_L = 0$. Momentum becomes

$$u_t + uu_x + gh_x = \frac{1}{9t}(-2\xi + 4c_L - 2\varphi) = 0, \quad (10)$$

because $\varphi = 2c_L - \xi$. The cancellation uses only the local formulas, so it is unaffected by cutting the fan off at $c = c_m$.

At $\xi = -c_L$ these formulas return $(h_L, 0)$. At $\xi = u_m - c_m$ they return $c = c_m$ and $u = 2(c_L - c_m) = u_m$. The rarefaction therefore meets both neighbouring constant states continuously.

5. The middle state

On $u_m - c_m < \xi < s$ the fields are the constants (h_m, u_m) . Every derivative vanishes, so both equations reduce to $0 = 0$. The region is non-empty: the shock section shows $s > u_m - c_m$.

6. The shock

Across $\xi = s$ the state jumps from (h_m, u_m) to $(h_R, 0)$. The mass jump fixes the speed already used in the candidate,

$$s(h_m - h_R) = h_m u_m. \quad (11)$$

The momentum jump required by the conservation law is

$$sh_m u_m = h_m u_m^2 + \frac{g}{2}(h_m^2 - h_R^2). \quad (12)$$

From the mass jump, $s - u_m = u_m h_R / (h_m - h_R)$, so the momentum jump rearranges to

$$h_m u_m (s - u_m) = \frac{h_m h_R u_m^2}{h_m - h_R} = \frac{g}{2}(h_m - h_R)(h_m + h_R). \quad (13)$$

Clearing the denominator gives

$$u_m^2 = \frac{g(h_m - h_R)^2(h_m + h_R)}{2h_m h_R}, \quad (14)$$

which is the shock formula for u_m . Both Rankine–Hugoniot relations hold.

The same formula gives an explicit speed,

$$s = \sqrt{\frac{gh_m(h_m + h_R)}{2h_R}}. \quad (15)$$

It lies strictly between the right-going characteristic speeds on the two sides. First,

$$s^2 - c_R^2 = \frac{g}{2h_R}(h_m^2 + h_m h_R - 2h_R^2) = \frac{g(h_m - h_R)(h_m + 2h_R)}{2h_R} > 0, \quad (16)$$

so $s > c_R = u_R + c_R$. Second,

$$(u_m + c_m) - s = c_m - \sqrt{\frac{gh_R(h_m + h_R)}{2h_m}}, \quad (17)$$

and the square of the first term exceeds the square of the second by

$$gh_m - \frac{gh_R(h_m + h_R)}{2h_m} = \frac{g(h_m - h_R)(2h_m + h_R)}{2h_m} > 0. \quad (18)$$

Thus $u_m + c_m > s > c_R$. Characteristics of the right-going family run into the shock, so the jump is an admissible shock. The same comparison yields $s - (u_m - c_m) = s - u_m + c_m > 0$, so the shock stays ahead of the rarefaction and the middle state occupies a strip of positive width.

Inside each open region the differential equations hold, the rarefaction joins the two constant states continuously, and the remaining jump satisfies the Rankine–Hugoniot relations together with the Lax condition. Stoker’s formula is a solution.